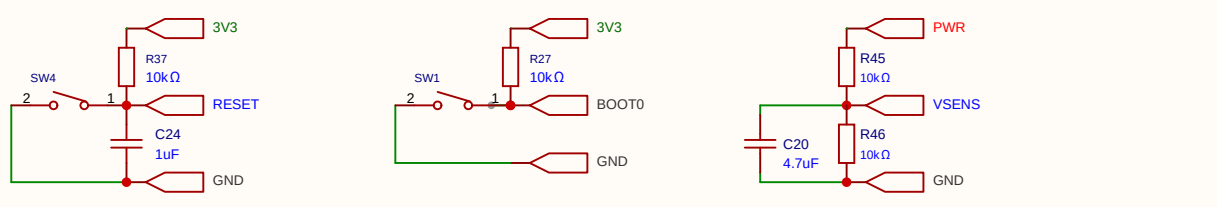
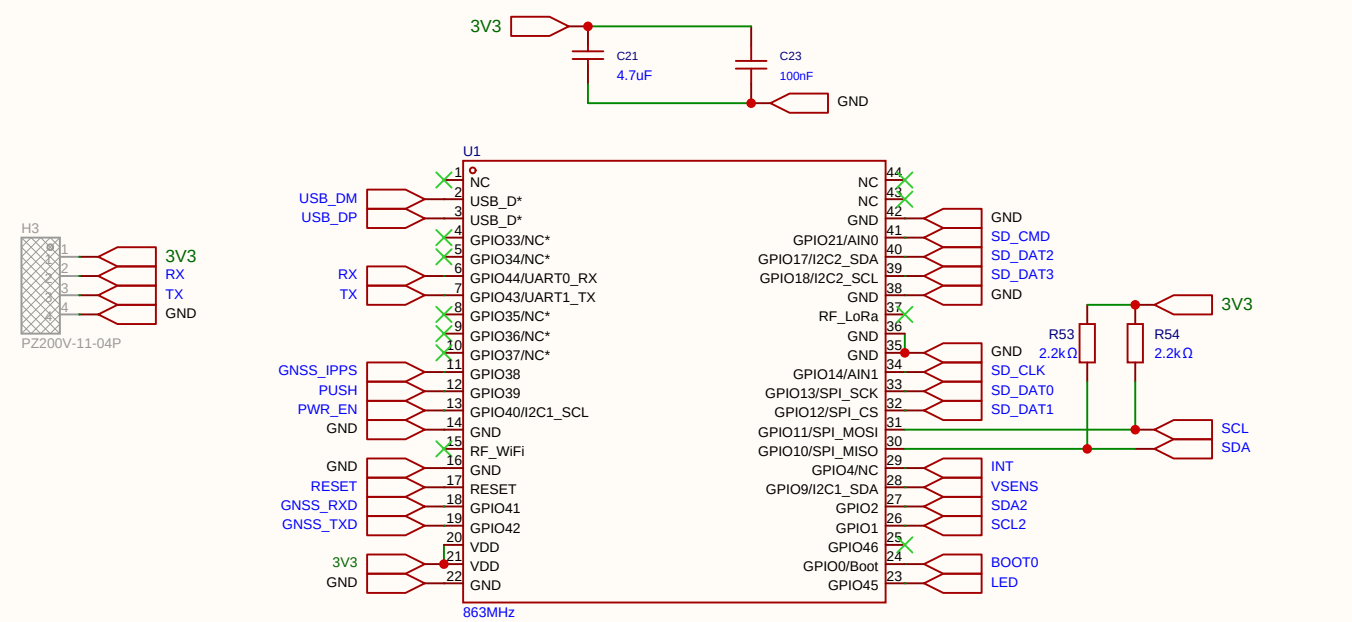
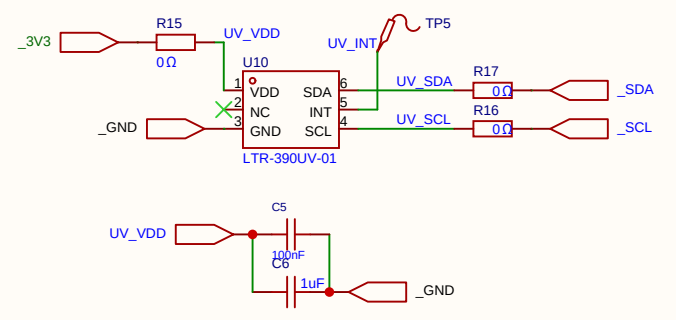


RAK3112 MCU+LORA Module

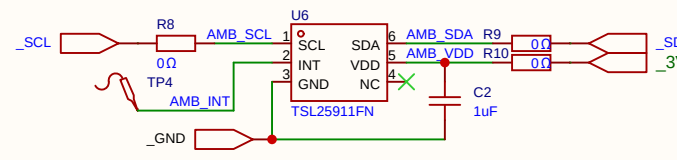


UV Sensor



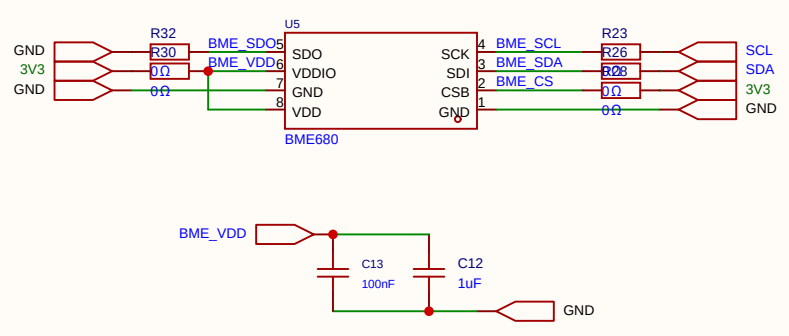
Pin	I/O Type	Symbol	Description
1	VDD		Power Supply Voltage
2	NC		No connection to this pin
3	GND		Ground
4	I	SCL*	I ² C serial clock. This pin is an open drain input.
5	O	INT*	Level Interrupt Pin. This pin is an open drain output.
6	I/O	SDA*	I ² C serial data. This pin is an open drain input / output.

Ambient Light Sensor



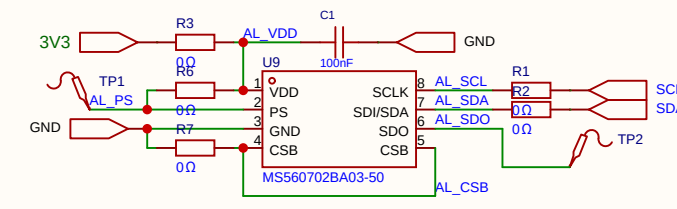
Pin Number	Pin Name	Description
1	SCL	I ² C serial clock input terminal
2	INT	Interrupt — open drain output (active low).
3	GND	Power supply ground. All voltages are referenced to GND.
4	NC	No connect — do not connect.
5	V _{DD}	Supply voltage
6	SDA	I ² C serial data I/O terminal

Humidity, Temperature Pressure Sensor



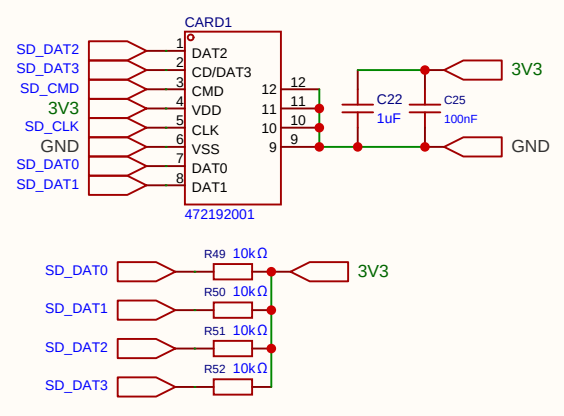
Pin	Name	I/O type	Description	Connection		
				SPI 4W	SPI 3W	I ² C
1	GND	Supply	Ground			
2	CSB	In	Chip select	CSB	CSB	V _{DDIO}
3	SDI	In/Out	Serial data input	SDI	SDI/SDO	SDA
4	SCK	In	Serial clock input	SCK	SCK	SCL
5	SDO	In/Out	Serial data output	SDO	DNC	GND for default address
6	V _{DDIO}	Supply	Digital / Interface supply	V _{DDIO}		
7	GND	Supply	Ground			
8	V _{DD}	Supply	Analog supply			V _{DD}

Altitude sensor



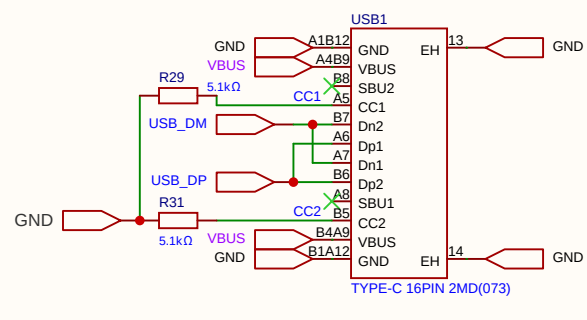
Pin	Name	Type	Function
1	VDD	P	Positive supply voltage
2	PS	I	Protocol select PS high (VDD) → I ² C PS low (GND) → SPI
3	GND	G	Ground
4	CSB	I	Chip select (active low), internal connection
5	SDO	O	Serial data output
6	SDI / SDA	I / IO	Serial data input / I ² C data IO
7	SCL	I	Serial data clock

SD Card

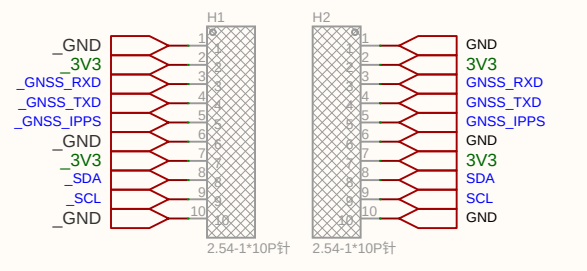
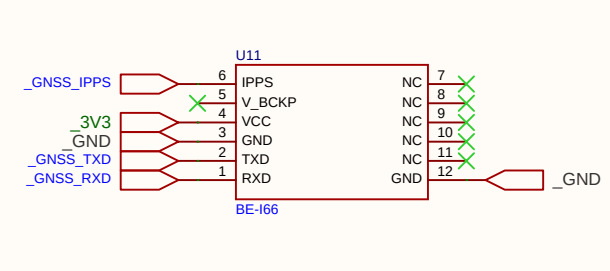


USB-C Connector

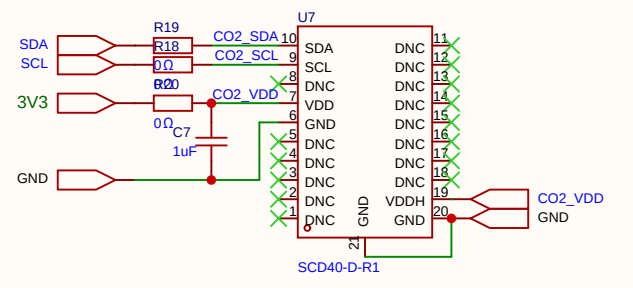
For Charging & Programming



GNSS receiver

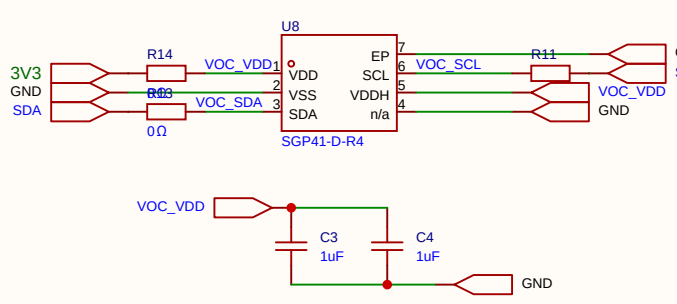


CO2 sensor



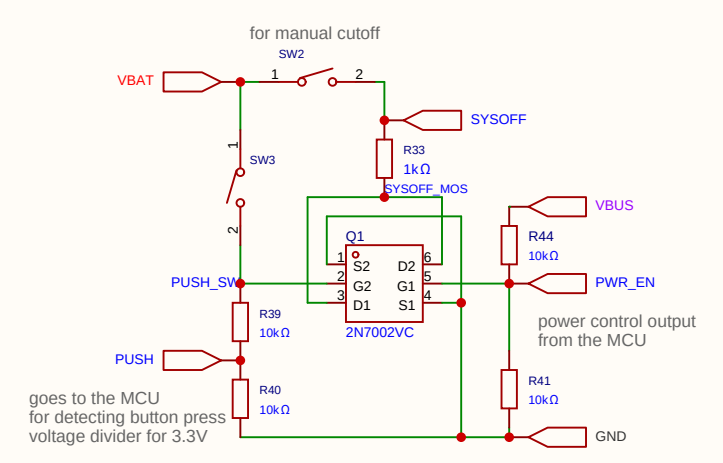
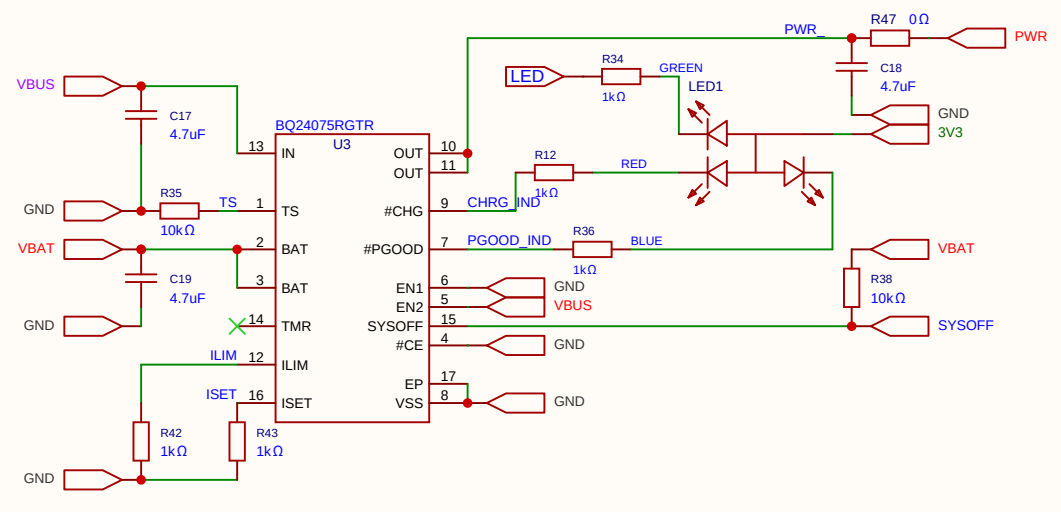
Name	Comments
VDD	Supply voltage
VDDH	Supply voltage IR source, must be connected to VDD on customer PCB
GND	Ground contact
SDA	I ² C Serial data, bidirectional
SCL	I ² C Serial clock
DNC	Do not connect, pads must be soldered to a floating pad on the customer PCB

VOC & NOx sensor

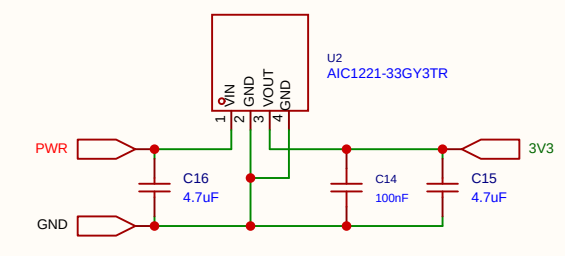


Pin	Name	Comments
1	VDD	Supply voltage
2	VSS	Ground
3	SDA	Serial data, bidirectional
4	n/a	Connect to ground (no electrical function)
5	VDDH	Supply voltage, hotplate
6	SCL	Serial clock, bidirectional

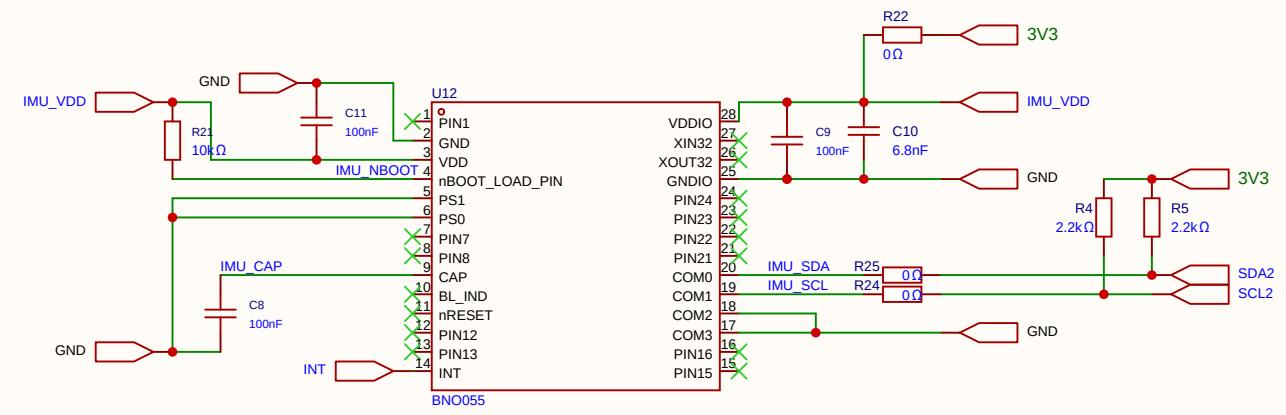
Charger & Power Control circuit



Voltage regulator



9-axis IMU sensor



Schematic	Schematic1	Create at	2026-07-29
Board	MAINBOARD	Update at	2026-09-27
Drawn		Page	P1
Reviewed			
Rocket BlackBox			
Version		Size	Page 1 Total 1
V1.0		A4	EasyEDA.com